

Model-Based Learning of Whittle indices

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Topics in Reinforcement Learning

Brief Idea of the Paper

Goal: Learn Whittle indices when (P, r) are unknown

Model-Based Approach

- Estimate transition matrices $\hat{P}$ and rewards $\hat{r}$
- Use structure of MDP $\rightarrow$ more sample efficient

Naive Idea

- Compute indices using WISC on $(\hat{P}, \hat{r})$

Problem

- Estimated MDP may **not be indexable**
- Indexability is **not robust** to small errors

Proposed Solution

- **EWISC:** Computes indices even for non-indexable MDPs
- **BLINQ:** Learning framework using EWISC

Non-Indexable Estimates Exist

Claim

- The set of indexable MDPs is **not an open set**. Small perturbations in (P, r) can break indexability.

Theorem

- There exists an MDP $M = (S, A, (P_a)_{a \in A}, (r_a)_{a \in A})$ that is **unichain and indexable**
- Such that $\forall \varepsilon > 0$, there exists a **unichain, non-indexable** MDP $\hat{M} = (S, A, (\hat{P}_a)_{a \in A}, (r_a)_{a \in A})$ with $\|M - \hat{M}\|_\infty \leq \varepsilon$

Proof Idea (from paper)

- Construct a family of MDPs parameterized by ε
- At $\varepsilon = 0$: MDP is **indexable**
- For any $\varepsilon > 0$: MDP becomes **non-indexable**

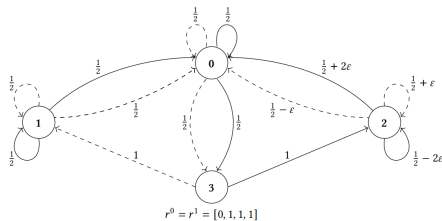


Figure: Counterexample MDP

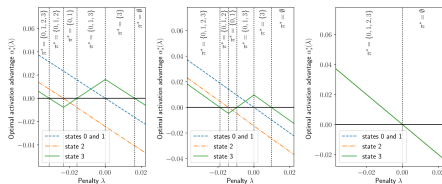


Fig. 3. In $\mathcal{M}_\epsilon$, activation advantage of each state under BO policies as λ varies, for $\epsilon = 0.01, 0.006$ and 0 , respectively.

Figure: Corresponding activation advantages

Basic Notation: Bias and Optimality

Bias Equation (under policy π)

$$g_s^\pi + b_s^\pi = r_s^\pi + \sum_{s' \in S} P_{s,s'}^\pi b_{s'}^\pi$$

Gain

$$g_s^\pi := \lim_{t \rightarrow \infty} \mathbb{E}[r^\pi(S_t) \mid S_0 = s]$$

Bellman Optimal (BO) Policy

$$g^* + b_s = \max_{a \in A} \left(r_s^a + \sum_{s'} P_{s,s'}^a b_{s'} \right)$$

Key Idea

- b = bias (relative value function)
- BO policy = optimal policy under average reward

Activation Advantage and Whittle Index

Activation Advantage

$$\alpha_s^\pi = r_s^1 - r_s^0 + (P_{s,\cdot}^1 - P_{s,\cdot}^0) \cdot b^\pi$$

Interpretation

- $\alpha_s^\pi > 0 \rightarrow$ action 1 is better
- $\alpha_s^\pi < 0 \rightarrow$ action 0 is better

Characterization of BO policy

- π is BO $\iff$
 - $\alpha_s^\pi \geq 0$ for $s \in \pi$
 - $\alpha_s^\pi \leq 0$ for $s \notin \pi$

Whittle Index Idea

- Define $\alpha_s^*(\lambda)$ under optimal policy
- Index λ_s satisfies:

$$\alpha_s^*(\lambda_s) = 0$$

WISC = Whittle Index in SubCubic time

Purpose: SOTA to compute Whittle indices for a given indexable MDP (P, r) .

Algorithm 1 Given a n -state arm, test indexability and compute Whittle index (if indexable).

```
1: Set  $\pi^1 := [n]$ ,  $\mu_{\min}^0 := -\infty$ 
2: if  $\pi^1$  is multichain then
3:   return the arm is multichain
4: end if
5: for  $k = 1$  to  $n$  do
6:   Compute  $\alpha^{\pi^k}(\lambda)$ 
7:   Let  $\mu_{\min}^k := \inf\{\lambda \geq \mu_{\min}^{k-1} : \exists i \in \pi^k, \alpha_i^{\pi^k}(\lambda) = 0\}$ 
8:   if  $\mu_{\min}^{k-1} < \mu_{\min}^k$  and for some  $i \notin \pi^k$ ,  $\alpha_i^{\pi^k}(\mu_{\min}^k) \geq 0$  then
9:     return the arm is not indexable
10:  end if
11:  if  $\mu_{\min}^k = +\infty$  then
12:    Set  $\lambda_i := +\infty$  for all  $i \in \pi^k$ 
13:    return the arm is indexable and the indices are  $\{\lambda_i\}_{i \in [n]}$ .
14:  end if
15:  Let  $\sigma^k \in \pi^k$  be such that  $\alpha_{\sigma^k}^{\pi^k}(\mu_{\min}^k) = 0$  and  $\lambda_{\sigma^k} = \mu_{\min}^k$ 
16:  Set  $\pi^{k+1} := \pi^k \setminus \{\sigma^k\}$ 
17:  if  $\pi^{k+1}$  is multichain then
18:    return the arm is multichain
19:  end if
20: end for
21: return the arm is indexable and the indices are  $\{\lambda_i\}_{i \in [n]}$ .
```

Figure: Pseudocode WISC

Why Do We Need EWISC?

Problem

- WISC works only if the MDP is **indexable**
- But estimated MDP $\hat{M}$ may be **non-indexable** (Based on the Theorem we saw earlier)

Goal

- Design an algorithm that:
 - Works for **any MDP**
 - Still approximates true Whittle indices

Solution: EWISC (Extended WISC)

Key Idea of EWISC

Recall

- WISC: finds the **unique** λ where $\alpha_s^*(\lambda) = 0$

Problem

- In non-indexable MDPs:
 - $\alpha_s^*(\lambda)$ may cross zero **multiple times**

EWISC Idea

- For each state s :
 - Collect **all** λ such that $\alpha_s^*(\lambda) = 0$
 - Return their **average**

Key Property

- If MDP is indexable $\Rightarrow$ only one crossing
- $\Rightarrow$ EWISC = WISC

What Does EWISC Compute?

High level description.

- Given an input MDP M , EWISC computes a finite series of $K \geq S$ penalties $\mu_0, \dots, \mu_K$ and $K + 1$ policies $\pi_0, \dots, \pi_K$ such that:
 - (1) (Policies “monotony”) $\pi_0 = S$, $\pi_K = \emptyset$
 - (2) (Penalties monotony) $\forall 1 \leq i \leq K - 1$, $\mu_i \leq \mu_{i+1}$
 - (3) (Optimality) Defining $\mu_0 := -\infty$ and $\mu_{K+1} := +\infty$, then $\forall 0 \leq i \leq S$ and $\forall \lambda \in (\mu_i, \mu_{i+1})$, π_i is the only BO policy in $M(\lambda)$

Difference from WISC

- $K \geq S$ (not exactly S)
- Policies need not evolve monotonically

EWISC: High-Level Intuition

Initialization

- $\lambda = -\infty$
- All states active: $\pi_0 = S$

Main Idea

- Increase λ
- Track when some state becomes indifferent:

$$\alpha_s^{\pi_i}(\lambda) = 0$$

At each step

- Find state σ whose advantage hits 0 first
- Update policy:
 - Flip σ (active $\leftrightarrow$ passive)

Repeat until

- $\pi = \emptyset$ (all states passive)

Algorithm 1 Extended WISC

Input: a MDP $\widehat{\mathcal{M}}$

Process: EWISC computes indices $-\infty = \mu_0 \leq \dots \leq \mu_{K+1} = +\infty$ and BO policies $\mathcal{S} = \pi_0 \supseteq \dots \supseteq \pi_K = \emptyset$ as described in §4.1

Output: for each $s \in \mathcal{S}$, the average of all penalties $\mu \in \mathbb{R}$ s.t. $\alpha_s^*(\mu) = 0$.

$\mu_0 \leftarrow -\infty, \pi_0 \leftarrow \mathcal{S}$

$i \leftarrow 0$

$\text{buff} \leftarrow \emptyset$

► Initialise buffer

while $\pi_i \neq \emptyset$ **do**

 Compute $\sigma \leftarrow \operatorname{argmin}_{s \in \mathcal{S} \setminus \text{buff}} \{\lambda \geq \mu_i \mid \alpha_s^{\pi_i}(\lambda) = 0\}$

$\mu \leftarrow \min\{\lambda \geq \mu_i \mid \alpha_s^{\pi_i}(\lambda) = 0\}$

if $\mu = \mu_i$ **then**

$\text{buff} \leftarrow \text{buff} \cup \{\sigma\}$

► Update buffer

else

$\text{buff} \leftarrow \{\sigma\}$

► Reset buffer

end if

$\mu_{i+1} \leftarrow \mu$

► Update indices sequence

$\pi_{i+1} \leftarrow \pi_i \Delta \{\sigma\}$

► By definition, $A \Delta B := (A \cup B) \setminus (A \cap B)$

$\pi_{i+1} \leftarrow \pi_i \cup \{s \in \mathcal{S} \setminus \pi_i \mid \alpha_s^{\pi_i}(\mu) > 0\}$

► Update policies sequence

$i \leftarrow i + 1$

end while

for each state $s \in \mathcal{S}$ **do**

$\text{temp} := \{\mu_i \mid 1 \leq i \leq K \text{ and } s \in \pi_{i-1} \Delta \pi_i\}$

$\text{out}[s] := \text{average}(\text{temp})$

► $\text{out}[s]$ contains the average of intersections of α_s^* with the x-axis

end for

return the vector out

Key Guarantee of EWISC

Property

- If $\hat{M} \rightarrow M$ (estimated model converges)

Then

- $\text{EWISC}(\hat{M}) \rightarrow$ true Whittle indices of M

Proof Idea

- $\forall s \in S, \quad \left| \text{EWISC}(\hat{M})_s - \lambda_s \right| \leq c_M \|M - \hat{M}\|_\infty$
- Small model error $\Rightarrow$ activation advantages are close:

$$\|\hat{\alpha}^*(\lambda) - \alpha^*(\lambda)\|_\infty \text{ is small}$$

- Indices = roots of $\alpha_s^*(\lambda) = 0$
- Close functions $\Rightarrow$ roots are close
- Even if multiple crossings exist, their average stays close

BLINQ: BLINQ Learning Is Not Q-learning

Idea

- Use EWISC to learn Whittle indices from data

Setting

- Unknown MDP M
- Access to a simulator (interaction-based learning)

Approach

- Learn an estimate $\hat{M}_t$
- Run EWISC on $\hat{M}_t$

Key Idea

- Since $\text{EWISC}(\hat{M}_t) \rightarrow \text{true indices}$, learning $\hat{M}_t$ is sufficient

Interaction

- At time t , observe state S_t
- Choose action $A_t \in \{0, 1\}$
- Observe reward and next state S_{t+1}

Counts

- $N_t(s, a)$: visits to (s, a)
- $N_t(s, a, s')$: transitions to s'

Transition Estimate

$$\hat{P}_{s,s'}^a(t) = \frac{N_t(s, a, s')}{N_t(s, a)}$$

Estimated MDP

$$\hat{M}_t = (S, A, \hat{P}_t, r)$$

Assumption

- Rewards are known and deterministic

Why?

- Learning rewards is easy
- Does not affect core difficulty

Focus

- Learn transition probabilities P_0, P_1

Algorithm 2 : BLINQ Learning Is Not Q-learning

At time $t = 0$:

$N_0(a, s) \leftarrow 0$ for every state $s \in \mathcal{S}$ and action $a \in \mathcal{A}$

$N_0(a, s, s') \leftarrow 0$ for every states $s, s' \in \mathcal{S}$ and action $a \in \mathcal{A}$

Start at some state S_0

End

At time $t \geq 1$:

Choose an action A_t uniformly in $\{0, 1\}$

Apply action A_t and transition to a new state S_{t+1} while earning a reward R_t

$N_{t+1}(S_t, A_t) \leftarrow N_{t+1}(S_t, A_t) + 1$

$N_{t+1}(S_t, A_t, S_{t+1}) \leftarrow N_{t+1}(S_t, A_t, S_{t+1}) + 1$

if $N_t(s, a) > 0$ for every $s \in \mathcal{S}, a \in \mathcal{A}$ and `update_indices(t) = True` **then:**

Build approximate MDP $\widehat{\mathcal{M}}_t$ as in Equation (8) and run EWISC($\widehat{\mathcal{M}}_t$)

end if

End

Time & Space Complexity

BLINQ

- Per step update: $O(1)$
- EWISC run: $O(KS^2)$ (with $K \geq S$, typically $O(S)$)
- EWISC runs: $f(T) = O(\log T)$ (spaced execution)
- **Total time:**

$$O(T) + O(S^3 \log T)$$

QWI / QGI

- Per step update (all states): $O(S)$
- **Total time:**

$$O(S \cdot T)$$

Space Complexity (all methods)

- $O(S^2)$ (transition matrices / value tables)

Key Insight

- BLINQ is more efficient in long run (EWISC run rarely)
- QWI/QGI require fine-tuned parameters; BLINQ is parameter-free

Experiments

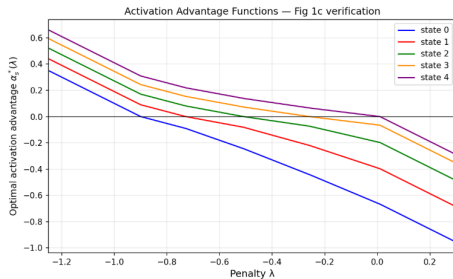


Figure: Whittle indices by EWISC

Experiments

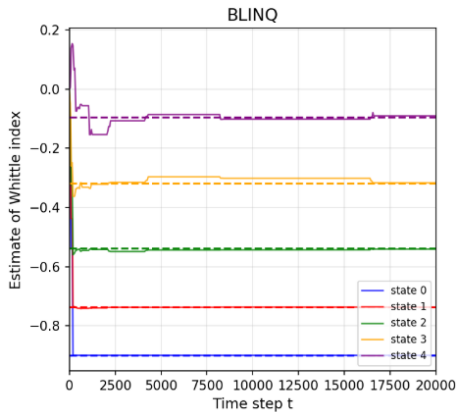


Figure: Experiment 6.1

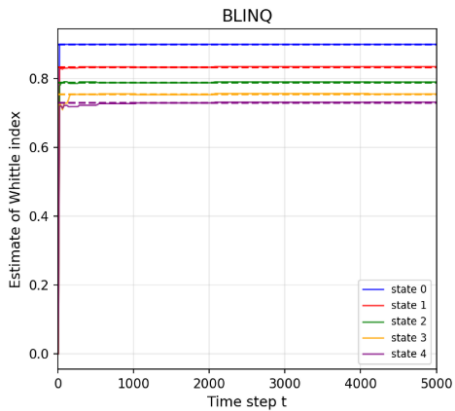


Figure: Experiment 6.2

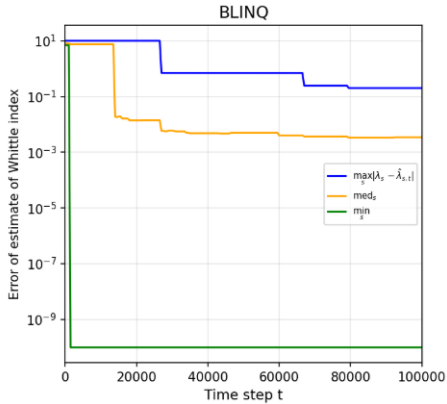


Figure: Experiment 6.3